Assignment 6: Hardwired Control Unit Design

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Deadline: Friday, 21 August 2026, 11:59 PM — exactly *two weeks* from issue (07 August 2026). There is no automatic late penalty — if you need more time, ask before the deadline and an extension will be granted (see the course policy in the syllabus).
Answer every question. Show all working for Numerical and Design questions.

6.1 Conceptual

Q1. Explain why every control signal in a hardwired control unit depends on *both* an opcode line (D_i) and a step line (T_j) together, rather than on either alone. Use Y_{in} as a concrete example in your explanation.

Q2. Explain what would happen to the control unit's behavior if the SC_{clear} signal were never wired up. Your answer must describe the effect on `SC`, on the step decoder's output, and on whether the CPU can execute more than one instruction.

6.2 Numerical

Q3. A CPU's instruction set contains 12 distinct instructions. What is the minimum number of one-hot opcode-decoder output lines (D_i) needed to distinguish them, and what is the minimum number of bits the `IR`'s opcode field must have to encode 12 instructions?

Q4. A sequence counter `SC` is implemented with 5 bits. What is the maximum number of distinct control steps (T_i states) this `SC` can address?

6.3 Design

Q5. Build the state table and derive the Boolean control equations (in the same style as $Y_{\text{in}} = D_{\text{ADD}} \cdot T_1$) for the instruction `R4` $\leftarrow$ `R5` $-$ `R6` on the single-bus datapath, assuming it has the same three-step shape as `ADD` but with the ALU set to subtract at T_2. Give one equation for each control signal in your state table.

Q6. Suppose a new instruction, `LOAD R1, (R2)` (register-indirect addressing), is added to the ISA, taking two control steps: T_1: MAR $\leftarrow$ [R2]; T_2: $R1 \leftarrow M[\text{MAR}]$. Write this instruction's state table, then draw (or describe precisely, gate by gate) the AND-OR realization of the control signal MAR_{in}, following the same method used in the lecture for Z_{in}.